

Performance Comparison of QKD Key Relaying Strategies Across Diverse Network Topologies

María Álvarez Roa^(1,*), Sebastian Verschoor⁽²⁾, Simon Rommel⁽¹⁾

⁽¹⁾ Department of Electrical Engineering, Eindhoven University of Technology, Eindhoven, Netherlands

⁽²⁾ Informatics Institute, University of Amsterdam, Amsterdam, Netherlands

^(*) m.alvarez.roa@tue.nl

Abstract *By evaluating across diverse network topologies, we provide a comparison of on-demand relaying and key flooding strategies for QKD using the key request success rate metric. Results show that key flooding outperforms on-demand relaying in sparse, low-connectivity networks, while on-demand relaying benefits from higher connectivity. ©2026 The Author(s)*

Introduction

The “store now and decrypt later” threat has accelerated the development of quantum-resilient communications such as Quantum Key Distribution (QKD), which enables two authenticated parties to exchange secret keys offering information-theoretic security (ITS) [1]. However, QKD is limited by link distance and low key generation rates, creating key challenges for deploying QKD networks (QKDNs). To address the distance-rate limitation, QKDNs with trusted nodes have been introduced as a practical solution until quantum repeaters become viable [2]. These networks extend secure communication over long distances through key relaying, a process coordinated by a key management system (KMS). In this approach, key material generated by QKD modules and managed by the KMS is consumed along the relay path to establish end-to-end secure keys.

In this paper, we analyze the scalability of two key distribution strategies across randomly generated network topologies with realistic parameters [3], using the key request success rate as the main performance metric. Simulations are performed for QKDNs employing on-demand relaying based on a hop-by-hop strategy and with the key flooding protocol [4, 5], which proactively distributes quantum keys across the network in a decentralized, flooding-like manner. While both approaches have been previously studied, their scalability with respect to network size has not been analyzed, and existing evaluations of the flooding protocol are limited to a single network topology. We address these gaps by studying both strategies across networks of varying sizes, densities and topologies. Results show that the success rate improves with increasing network density and average node degree, with performance differences driven by network connectivity. Key flooding performs better in sparse networks, while on-demand relaying becomes more effective as connectivity increases.

QKD Network and Key Relaying Strategies

The simulated QKDN architecture follows the standardized model defined in [6]. Each simulated node has one KMS which may interface with multiple co-located QKD modules. For every connected pair of QKD modules, key material is generated and periodically provided to the corresponding local KMSs. The KMSs are responsible for storing the generated key material for relay purposes and for handling key requests from client applications. To prevent the risk of key reuse caused by potential synchronization issues, the shared key material between each pair of KMSs is divided into two where one half was used for key relaying in one direction and the other half for key relaying in the other direction [7]. Additionally, balancing of key pool levels is considered in cases where buffer utilization becomes asymmetric.

In practice, different strategies are used to distribute and establish key material between end nodes, and to control how it is utilized. The two main approaches are on-demand relaying and proactive relaying. In the on-demand approach, key relay is triggered only upon request, typically via a hop-by-hop procedure [8]. At each QKD node, the message authenticity is verified and the keys are decrypted and re-encrypted at every hop until reaching the destination. Keys are protected with one-time pad (OTP) encryption, and each message is authenticated using an ITS message authentication scheme. This approach improves efficiency by consuming keys only when needed but can introduce additional latency due to the full relay process taking place only upon request rather than ahead of time. The implemented protocol selects relay routes from multiple shortest paths while considering the local key pool status of the first hop. Alternatively, extending this on-demand approach, the ITU defines a scheme where XOR-combined adjacent link keys are forwarded to a central node [6], which derives the final key for the destination. Although not strictly

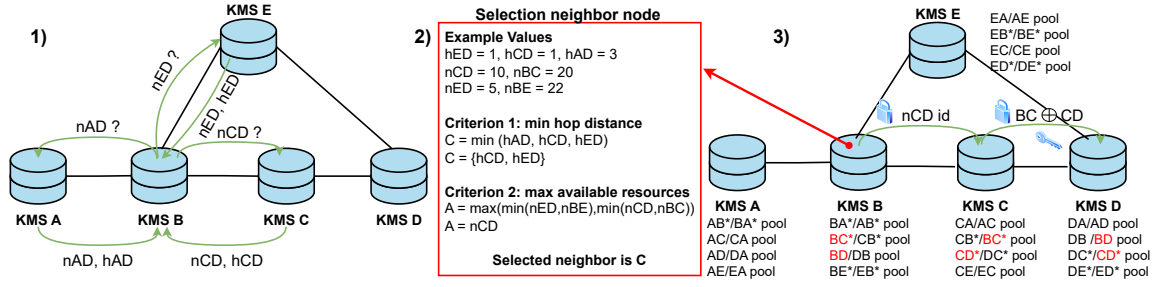


Fig. 1: Key distribution in key flooding protocol. OTP encryption ($\oplus$) protects distributed keys and the padlock denotes ITS message authentication. In the example, the pool to fill is BD with KMS B the initiator. In step 1), KMS B requests from each adjacent KMS the number of keys available to reach D. In step 2) it selects the one with the lowest number of hops (h) and higher key availability (n). Finally, in step 3), the actual key distribution is carried out. In the example, key pools used during relaying are shown in red.

hop-by-hop, this approach follows the same principle and is expected to exhibit similar performance.

In contrast, proactive relaying aims to minimize service degradation by distributing key material in advance from adjacent to non-adjacent key pools across the network. By pre-establishing end-to-end key pools, keys can be delivered immediately when requested. Each QKDN node manages its own pool and coordinates with neighbors for efficient distribution. This improves response time and service quality, but its effectiveness depends on how well the pre-established pools match actual demand. It may also increase key consumption when smaller keys are relayed. The implemented protocol follows the key flooding approach [4, 5] and includes the enhancements from [9, enhancement 1b-2-3]. To fill all key pools in the network, each QKD node periodically checks which of its non-adjacent key pools need keys. Once the pool to be filled is selected, the key relaying process starts, as shown in Fig. 1.

Topology Generation and Simulation Design

To model the QKD network, a set of topologies is generated to evaluate the performance of the proposed key distribution strategies. Using the parameters defined in Table 1, the NetworkX library [10] is employed to create connected graphs with a specified number of nodes and links. The generation process enforces sanity checks, including connectivity, a minimum node degree of at least 2, and bounded average degree and degree dispersion. From each topologies, metrics such as network density (ρ) and average node degree (deg) are computed to characterize network size and connectivity, critical for analyzing the scalability of the strategies under study. For a fixed deg , ρ is controlled via N and L ; lower ρ yields more constrained link distributions and fewer routing paths. The dataset [3] also includes statistics (e.g., hop-distance distributions) and topology visualizations.

The simulator that models and evaluates the management of key material in a QKDN is based on discrete-event simulation prioritizing speed, flexibility, and scalability, enabling rapid exploration

Tab. 1: Topology parameters

Parameter	Description
N	Number of nodes
L	Number of links
$\rho = \frac{2L}{N(N-1)}$	Network density
$deg = \frac{2L}{N}$	Average node degree
$\sigma_{deg}^{max} = \frac{deg}{2}$	Maximum standard deviation of the node degree

Tab. 2: Simulation parameters

Parameter	Values
Average inter-request interval	2 s
Inter-arrival time of key material	40 s
Secret key rate	10 kbyte/s
Average bytes requested	1280 byte
Max Capacity	50 kbyte
Client key reserve	1500 byte
Flooding trigger interval	20 s

of alternative key relaying strategies and the evaluation of many and large network topologies. It models the dynamics of key generation, storage, and consumption, providing a detailed KMS representation. The validity of the proposed simulation is verified via a Docker-based deployment using an in-house developed fully functional KMS as a reference implementation.

Table 2 summarizes the relevant system parameters affecting performance. For simplicity, all key pools are assumed to have the same maximum capacity. Client key requests arrive according to an exponential distribution with a mean inter-arrival time of 2 s. Request sizes follow a uniform distribution between 960 and 1600 bytes. In terms of key generation, quantum links are modeled with fixed secret key rates to enable fair comparison across network topologies and isolate the effect of network structure from physical-layer variability. Key material is periodically provided to the KMS with an inter-arrival time of 40 s. In the key flooding strategy, we fix the flooding trigger interval, which controls how often non-adjacent key pools are refilled. The client key reserve refers to the minimum

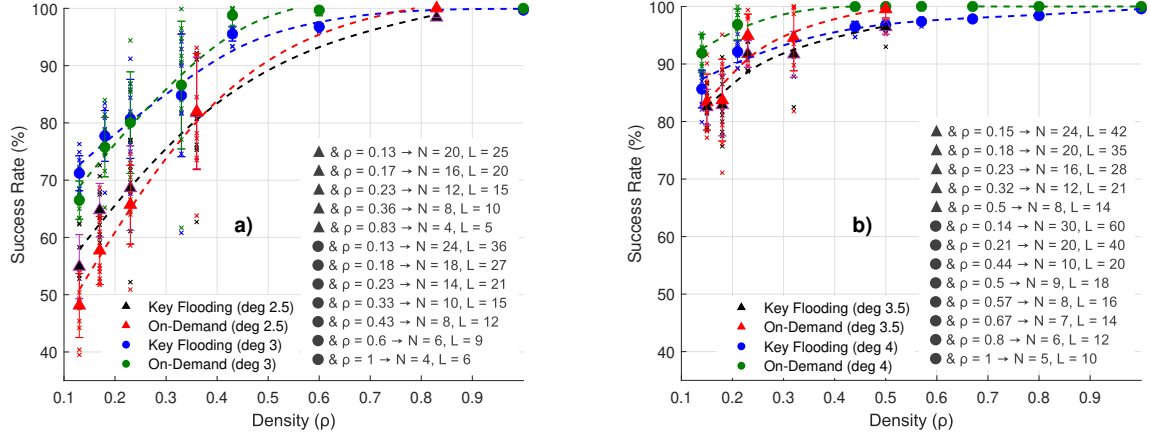


Fig. 2: Overall success rate vs network density for key flooding and on-demand relaying. a) and b) show results for different average node degrees (2.5, 3, 3.5, and 4), with triangles and circles distinguishing values. Markers denote mean values, scattered points show single topologies, and curves are smoothing spline fits for visualization.

number of keys for clients allocated to each key pool. Once this minimum target is achieved, surplus keys can be used for relaying and authentication. Simulations are event-driven and span 7200 seconds per run, with a mean runtime of ~ 2 minutes. Starting from empty pools, steady state is reached after ~ 1800 seconds and this is excluded from average performance metrics.

Simulation Results and Discussion

Fig. 2 shows the overall success rate (SR) of the network, defined as the proportion of requests that are successfully fulfilled, for different topologies characterized by their density (ρ) and average node degree (deg). The number of nodes and links in each topology associated to ρ and deg is included in Fig. 2. For each parameter configuration, 10 different network topologies were generated and individual as well as aggregate statistics are shown in Fig. 2. The SR is evaluated versus network density for key flooding and on-demand relaying. Results are presented for average node degrees of 2.5 and 3 (Fig. 2 a)) and 3.5 and 4 (Fig. 2 b)). Each marker represents the mean SR over multiple topologies with the same characteristics, with error bars indicating the standard deviation. Variations across topologies with the same characteristics arise due to network structure (e.g., maximum node hop count) [3]. To mitigate stochastic variability, multiple simulation runs were performed for each topology. Reported values are the average, represented as crosses.

The SR increases with both ρ and deg . Lower ρ restricts link distribution, which can limit paths between distant nodes. This affects on-demand relaying by requiring longer paths with more hops and higher key consumption, and key flooding by making end-to-end key pool establishment between distant nodes more difficult, as closer nodes are prioritized first. For example, at $deg = 3$, increasing the network density from $\rho = 0.13$ to

$\rho = 1$ raises the SR by 33.17% for on-demand relaying and 25.57% for key flooding.

When comparing networks with similar ρ , increasing deg improves SR by enhancing local connectivity and providing more direct neighbors, enabling shorter and more flexible routes for on-demand relaying, while also facilitating key pool establishment for key flooding. As shown in Fig. 2, this effect is more pronounced at low densities; for instance, for $\rho \in [0.13, 0.15]$ and deg from 2.5 to 4, SR increases by 43.78% for on-demand relaying and 30.75% for key flooding. Network density ranges from 0.1 to 1 (mesh network); missing values correspond to infeasible (N , L) combinations.

For low-density networks, key flooding outperforms on-demand relaying, especially for less connected networks (Fig. 2 a)). This is because on-demand requires a complete, resource-sufficient path between source and destination at the time of the relaying, which becomes increasingly unlikely in sparsely connected networks. However, when ρ increases on-demand has higher SR than key flooding. Note that based on the optimization in [5] a fixed client key reserve of 1500 bytes is used as a representative value. Further improvements for key flooding might be obtained by increasing this threshold, especially in higher-density topologies.

Conclusions

We evaluate the performance of on-demand relaying and key flooding strategies across diverse network topologies. Results show that both network density and average node degree significantly impact performance, highlighting the importance of connectivity in QKD network design. While key flooding outperforms hop-by-hop relaying in networks with low density and low average node degree, this is inverted for highly connected networks and higher densities, highlighting the importance of tailoring the selected key distribution strategy depending on the network conditions.

Acknowledgements

This work was in part supported by the Dutch Ministry of Economic Affairs and Climate Policy (EZK) as part of the Quantum Delta NL programme and the EU's Digital Europe Programme under project QCINed (grant no. 101091656) within the EuroQCI initiative.

References

- [1] C. H. Bennett and G. Brassard, "Quantum cryptography: Public key distribution and coin tossing", *Theoretical Computer Science*, vol. 560, pp. 7–11, Dec. 2014, ISSN: 0304-3975. DOI: 10.1016/j.tcs.2014.05.025. [Online]. Available: <http://dx.doi.org/10.1016/j.tcs.2014.05.025>.
- [2] Y. Cao, Y. Zhao, Q. Wang, J. Zhang, S. Ng, and L. Hanzo, "The Evolution of Quantum Key Distribution Networks: On the Road to the Qinternet", *IEEE Communications Surveys & Tutorials*, vol. 24, pp. 839–894, Jan. 2022. DOI: 10.1109/COMST.2022.3144219.
- [3] M. Alvarez Roa, S. Verschoor, and S. Rommel, *Generated Network Topologies for QKD Key Relaying Strategies Evaluation*, Zenodo, 2026. DOI: 10.5281/zenodo.19550389. [Online]. Available: <https://doi.org/10.5281/zenodo.19550389>.
- [4] M. Alvarez Roa, S. Verschoor, and S. Rommel, "Efficiency Analysis of Two Key Relaying Architectures for QKD Networks", in *ECOC 2024; 50th European Conference on Optical Communication*, 2024, pp. 1389–1392.
- [5] M. Alvarez Roa, C. Stan, S. Verschoor, I. Tafur Monroy, and S. Rommel, "Decentralized key distribution versus on-demand relaying for QKD networks", *Journal of Optical Communications and Networking*, vol. 17, no. 8, pp. 732–742, 2025. DOI: 10.1364/JOCN.547793.
- [6] ITU-T, "Quantum key distribution networks – Key management", International Telecommunication Union, Standard Y.3803, Dec. 2020. [Online]. Available: <https://www.itu.int/rec/T-REC-Y.3803-202012-I/en>.
- [7] T. van Duijn, S. R. Verschoor, and S. Rommel, "Synchronization Protocol for Quantum Key Distribution Networks", in *Advanced Photonics Congress (IPR, Networks, NOMA, SOLITH, SPPCom)*, Optica Publishing Group, 2025, NeM3C.4. DOI: 10.1364/NETWORKS.2025.NeM3C.4. [Online]. Available: <https://opg.optica.org/abstract.cfm?URI=Networks-2025-NeM3C.4>.
- [8] M. Peev, C. Pacher, R. Alléaume, and others., "The SECOQC quantum key distribution network in Vienna", *New Journal of Physics*, vol. 11, no. 7, p. 075 001, Jul. 2009. DOI: 10.1088/1367-2630/11/7/075001. [Online]. Available: <https://dx.doi.org/10.1088/1367-2630/11/7/075001>.
- [9] M. Álvarez Roa, S. Verschoor, and S. Rommel, "Enhancing Key Availability and Efficiency in QKDNs via Decentralized Key Flooding", in *30th International Conference on Optical Network Design and Modelling, ONDM 2026*, Munich, 2026.
- [10] A. Hagberg, P. Swart, and D. Chult, "Exploring Network Structure, Dynamics, and Function Using NetworkX", Jun. 2008. DOI: 10.25080/TCWV9851.